

Hierarchical Subspace Empirical Bayes Tournament: Temporal Topo-HDDM vs. Baselines

Cerebellar Cognitive Modeling Project

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1 Executive Summary

This document details the final mathematical formalization and rigorous out-of-sample evaluation of three generative cognitive models: Win-Stay-Lose-Shift (WSLS), Rescorla-Wagner Counterfactual (RW-CF), and the highly parameterized Temporal Topological HDDM.

Following severe computational bottlenecks with high-dimensional evolutionary sampling (CMA-ES), the methodology was radically upgraded to a **Manifold-Constrained Hierarchical Bayesian Model (Empirical Bayes)**. This allowed the 15-dimensional biological network to be strictly regularized via an empirical 6-dimensional subspace prior, allowing for blazing fast and perfectly converged out-of-sample testing using Precision-Recall Area Under the Curve (PR-AUC) on cognitive “switch” trials.

The definitive conclusion is that the **RW-CF model is the undisputed victor**, fundamentally outperforming both the Topo-HDDM and WSLS baselines.

2 Methodology

2.1 Model Specifications

1. **M1: WSLS (3 Parameters)**: A baseline heuristic predicting responses based purely on the previous outcome.
2. **M2: RW-CF (5 Parameters)**: A Rescorla-Wagner associative learning model with counterfactual updating and dynamic drift-rate scaling.
3. **M3: Temporal Topo-HDDM (15 Parameters)**: A highly recurrent cerebellar network mapping 10-dimensional temporal mossy fiber inputs (kinematics, RTs, time-to-feedback) onto a 500-cell granular manifold to dynamically modulate decision boundaries.

2.2 Manifold-Constrained Empirical Bayes

To combat the curse of dimensionality, an unconstrained CMA-ES extraction was performed on 10 subjects. Principal Component Analysis (PCA) revealed that **92.2% of the biological variance was mathematically constrained to a 6-dimensional subspace**.

The 10-fold Cross-Validation tournament utilized this structural prior:

- **E-Step**: Each training subject was optimized within the 6D latent manifold.
- **M-Step**: The population hyper-mean (μ_{latent}) and hyper-variance (σ_{latent}) were updated to enforce Bayesian shrinkage.
- **Out-of-Sample Test**: The fully converged population hyper-mean was used to predict the PR-AUC and Negative Log-Likelihood (NLL) of the held-out subject.

3 Results

The tournament was executed on 10 distinct participants. The out-of-sample predictive performance on cognitive switch trials is summarized below:

Model	Mean PR-AUC (Switch)	Mean Out-of-Sample NLL
M1: WSLS	0.504	147.2
M2: RW-CF	0.661	141.4
M3: Topo-HDDM	0.519	590.1

Table 1: Cross-Validated Predictive Performance across 10 held-out subjects. Higher PR-AUC and lower NLL indicate superior predictive power.

3.1 Conclusion

Despite resolving all mathematical and computational bottlenecks via Subspace Empirical Bayes—which successfully shrunk the Topo-HDDM parameters to a robust population mean—the model **fails to outperform simple associative learning**.

The 15-dimensional Temporal Topo-HDDM yields a PR-AUC of 0.519, behaving functionally identical to the naïve WSLS heuristic (0.504). Furthermore, its joint probability deviance (NLL $\approx$ 590) is significantly worse than both baselines.

The integration of granular topology and temporal mossy fibers does not yield any meaningful predictive state information regarding cognitive switching. The RW-CF model (PR-AUC 0.661) remains the absolute optimal generative architecture for this behavioral task.